Chapter 5. Association and Independence of Two Variables

Pearson and Spearman correlation, Kendall's tau, permutation tests of independence

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5.1 Pearson correlation

For a pair (X, Y), $\rho = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}\,X\,\mathrm{Var}\,Y} \in [-1,1]$ measures *linear* association. Independence implies $\rho = 0$; the converse holds under bivariate normality but not in general (e.g. $Y = X^2$ with X symmetric gives $\rho = 0$). From pairs (X_i, Y_i), $i = 1, \dots, n$, the sample correlation is $r = S_{XY}/\sqrt{S_{XX}S_{YY}}$. The least squares slope is $\hat\beta = S_{XY}/S_{XX} = r\sqrt{S_{YY}/S_{XX}}$. For the population, $\beta = \mathrm{Cov}(X,Y)/\mathrm{Var}\,X = \rho\,\sigma_Y/\sigma_X$, so testing $\beta = 0$ in a simple regression and testing $\rho = 0$ are the same problem.

Proposition 5.1

If (X_i, Y_i) are i.i.d. bivariate normal with $\rho = 0$, then $T = r\sqrt{(n-2)/(1-r^2)} \sim t_{n-2}$.

Proof. Condition on the X's. Then Y_i are i.i.d. normal, i.e. the regression model with $\beta = 0$, and the slope t-statistic $\hat\beta/\widehat{\mathrm{se}}(\hat\beta) \sim t_{n-2}$. With $SSE = S_{YY}(1-r^2)$ and $\widehat{\mathrm{se}}^2 = SSE/((n-2)S_{XX})$, one gets $\hat\beta/\widehat{\mathrm{se}} = r\sqrt{(n-2)/(1-r^2)}$. The conditional law does not depend on the X's. ■

Permutation test of independence. Under H_0: $X \perp Y$, the Y's can be permuted among the fixed X's without changing the joint law, so shuffling Y gives an exact test with any statistic, e.g. r. The alternative it is sensitive to is linear association. Note that H_0 is independence, not $\rho = 0$: if $\rho = 0$ but X, Y are dependent, shuffling is not justified.

Proposition 5.2

Over all $n!$ permutations of $(Y_1, \dots, Y_n)$ with the X's fixed, the permutation distribution of r has mean 0 and variance $1/(n-1)$, whatever the data.

Proof. Let $a_i = X_i - \bar X$, $b_i = Y_i - \bar Y$ and π uniform. Then $r = \sum_i a_i b_{\pi(i)}/\sqrt{S_{XX}S_{YY}}$ and $E b_{\pi(i)} = 0$. Also $\mathrm{Var}\,b_{\pi(i)} = S_{YY}/n$ and $\mathrm{Cov}(b_{\pi(i)}, b_{\pi(j)}) = -S_{YY}/(n(n-1))$ for $i \neq j$. With $\sum_{i\neq j} a_i a_j = -S_{XX}$,

$$\mathrm{Var}\left(\sum a_i b_{\pi(i)}\right) = \frac{S_{XX}S_{YY}}{n} + \frac{S_{XX}S_{YY}}{n(n-1)} = \frac{S_{XX}S_{YY}}{n-1}.$$

■

Combined with a finite-population CLT, $\sqrt{n-1}\,r$ is approximately $N(0,1)$ under the permutation distribution, a distribution-free counterpart of Proposition 5.1.

5.2 Spearman's rank correlation

Replace X_i, Y_i by their ranks R_i, S_i and compute the Pearson correlation: $r_s = \mathrm{Corr}(R, S)$. It measures *monotone* association, is invariant under increasing transformations of either variable, and is robust to outliers.

Proposition 5.3

Without ties, $r_s = 1 - \dfrac{6\sum_i (R_i - S_i)^2}{n(n^2-1)}$.

Proof. Both rank vectors are permutations of $1, \dots, n$, with mean $\frac{n+1}{2}$ and $\sum(R_i - \bar R)^2 = \frac{n(n^2-1)}{12}$. Since $\sum(R_i - S_i)^2 = \sum(R_i - \bar R)^2 + \sum(S_i - \bar S)^2 - 2\sum(R_i - \bar R)(S_i - \bar S)$, we get $\sum(R_i - \bar R)(S_i - \bar S) = \frac{n(n^2-1)}{12} - \frac{1}{2}\sum(R_i - S_i)^2$. Divide by $\frac{n(n^2-1)}{12}$. ■

The permutation test with r_s tests independence against monotone association. It is distribution-free, and Proposition 5.2 applies to it verbatim.

5.3 Kendall's tau

A pair of observations $i < j$ is **concordant** if $(X_i - X_j)(Y_i - Y_j) > 0$ and **discordant** if < 0. Let $U_{ij} = \operatorname{sgn}[(X_i - X_j)(Y_i - Y_j)]$.

Definition (Kendall's tau)

Sample: $\tau_n = \binom{n}{2}^{-1} \sum_{i<j} U_{ij} = (N_c - N_d)/\binom{n}{2}$. Population: for independent copies $(X, Y), (X', Y')$, $\tau = P\{(X - X')(Y - Y') > 0\} - P\{(X - X')(Y - Y') < 0\}$.

τ_n is unbiased for τ. Unlike r_s, it has a clear population counterpart.

Example 5.1

$x = (1, 2, 3, 4, 5)$, $y = (2, 1, 4, 3, 5)$. Only the pairs $(1, 2)$ and $(3, 4)$ are discordant, so $\tau_n = (8 - 2)/10 = 0.6$. The rank differences are $(-1, 1, -1, 1, 0)$, so by Proposition 5.3, $r_s = 1 - 6 \cdot 4/120 = 0.8$. Pearson's r is also 0.8 here, since the data are their own ranks.

Proposition 5.4 (Normal case)

If (X, Y) is bivariate normal with correlation ρ, then $\tau = \frac{2}{\pi} \arcsin \rho$.

Proof. $(U, V) = (X - X', Y - Y')$ is centered normal with correlation ρ. Write $U = Z_1$ and $V = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$, then pass to polar coordinates $(Z_1, Z_2) = R(\cos\Theta, \sin\Theta)$ with Θ uniform, and set $\rho = \sin\phi$. Then $V = R\sin(\Theta + \phi)$. The set where $\cos\Theta$ and $\sin(\Theta + \phi)$ share a sign has total angle $\pi + 2\phi$, so $P(UV > 0) = \frac{1}{2} + \frac{\phi}{\pi}$ and $\tau = 2P(UV > 0) - 1 = \frac{2\phi}{\pi}$. ∎

This gives a robust estimator $\hat{\rho} = \sin(\pi\tau_n/2)$ of a normal correlation in the presence of outliers.

Proposition 5.5

If $X \perp Y$ with continuous margins, then $E\tau_n = 0$ and $\operatorname{Var}\tau_n = \dfrac{2(2n + 5)}{9n(n - 1)}$. Moreover $\tau_n / \sqrt{\operatorname{Var}\tau_n} \xrightarrow{d} N(0, 1)$.

Proof. Each $U_{ij} = \pm 1$ with probability $\frac{1}{2}$, so $\operatorname{Var} U_{ij} = 1$. Pairs with no common index give independent U's. For a shared index, independence of X and Y gives $E(U_{12}U_{13}) = \left(E[\operatorname{sgn}(X_1 - X_2)\operatorname{sgn}(X_1 - X_3)]\right)^2 = \left(E(2V - 1)^2\right)^2 = \frac{1}{9}$ with V uniform. There are $\binom{n}{2} \cdot 2(n - 2)$ ordered pairs of pairs sharing one index, so

$$\operatorname{Var}\Big(\sum_{i<j} U_{ij}\Big) = \binom{n}{2}\Big(1 + \frac{2(n - 2)}{9}\Big) = \binom{n}{2}\frac{2n + 5}{9},$$

and dividing by $\binom{n}{2}^2$ gives the formula. Asymptotic normality follows from Hoeffding's theory of U-statistics. ∎

Under independence the exact null distribution of τ_n is its permutation distribution (shuffle Y), which is distribution-free. Without ties this is Kendall's tabulated null distribution. So there are three equivalent ways to test independence:

(i) a (random) permutation test;

(ii) Kendall's exact null distribution;

(iii) the normal approximation of Proposition 5.5, already good for $n \approx 10$ except in the extreme tails. Pairs with $(X_i - X_j)(Y_i - Y_j) = 0$ give $U_{ij} = 0$. Software usually reports the tie-corrected version $\tau_b = (N_c - N_d)/\sqrt{(N_0 - T_X)(N_0 - T_Y)}$, where $N_0 = \binom{n}{2}$ and T_X, T_Y count the pairs tied in X or in Y. With ties only the permutation test remains exact.

Which measure? Under bivariate normality, use the t-test of r. With heavy-tailed margins but linear dependence, rank-based r_s and τ_n give more powerful tests. For monotone nonlinear dependence, r_s or τ_n are the natural measures. All three can be near zero for strong non-monotone dependence (e.g. $Y = \sin(3X) + \epsilon$), so failing to reject independence does not mean independence.